aggregates of cells *in vitro* in liquid medium is known as cell suspension culture. The cell suspension is prepared by transferring a portion of callus to the liquid medium and agitated using rotary shaker instrument. The cells are separated from the callus tissue and used for cell suspension culture.

Production of Secondary Metabolites

Cell suspension culture can be useful for the production of secondary metabolites like alkaloids, flavonoids, terpenoids, phenolic compounds and recombinant proteins. Secondary metabolites are chemical compounds that are not required by the plant for normal growth and development but are produced in the plant as 'byproducts' of cell metabolism. For Example: Biosynthesis and isolation of indole alkaloids from *Catharanthus roseus* plant cell culture.

The process of production of secondary metabolites can be scaled up and automated using bio-reactors for commercial production. Many strategies such as biotransformation, elicitation and immobilization have been used to make cell suspension cultures more efficient in the production of secondary metabolites. Few examples of industrially important plant secondary metabolites are listed below in the table:

Secondary metabolites	Plant source	Uses
Digoxin	*Digitalis purpurea*	Cardiac tonic
Codeine	*Papaver somniferum*	Analgesic
Capsaicin	*Capsicum annuum*	Rheumatic pain treatment
Vincristine	*Catharanthus roseus*	Anti-carcinogenic
Quinine	*Cinchona officinalis*	Antimalarial

Table 5.1: Secondary metabolites and its plant resources

5.4 Plant Regeneration Pathway

From the explants, plants can be regenerated by somatic embryogenesis or organogenesis.

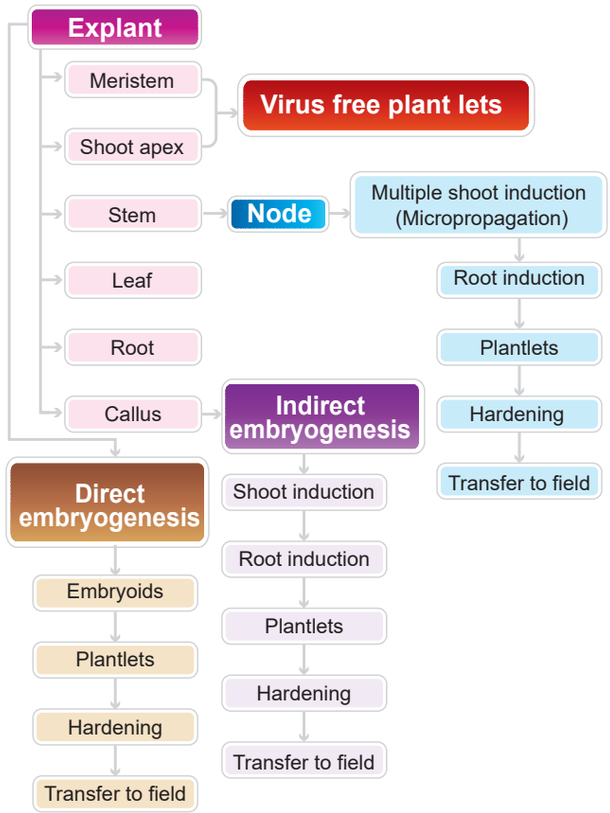


Figure 5.9: Flow chart of Plant regeneration pathway

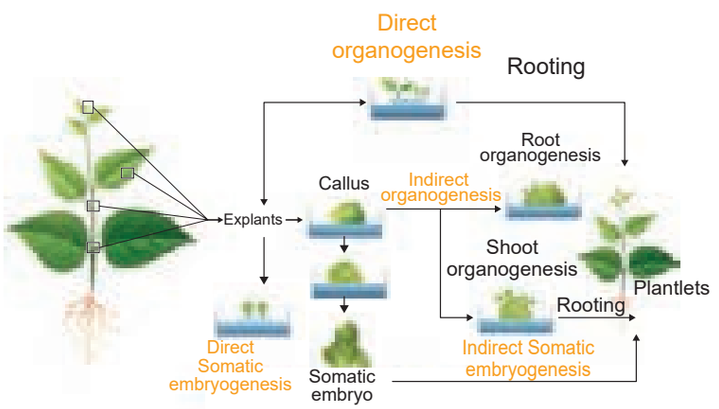


Figure 5.10: Plant Regeneration Pathway

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